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The importance of indicator weights for vulnerability indices and implications for decision making in disaster management

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ABSTRACT

Decision makers in the field of disaster management have many tools at their disposal to assist them in making informed decisions. They make use of the available information to set priorities, choose adequate local adaptation measures and make decisions regarding land use and emergency planning. As far as the vulnerability assessment of buildings is concerned, a wide range of methods can be found in the literature. However, some methods (e.g. vulnerability curves) require a large amount of empirical data that are not always available. An alternative solution is the development of vulnerability indices. In this paper, two relative vulnerability indices (PTVA-3 and PTVA-4) designed for the assessment of vulnerability of buildings to tsunamis are applied. The case study area is located in Apulia (Italy). Although no historic record of tsunamis is available for the area, geological investigations have shown that the area has been affected by tsunamis in the past. The two methods use different weighting for the indicators considered for the vulnerability index leading to different results. In this way, the importance of the weighting of indicators and its implication in decision making is shown. Moreover, the study offers the opportunity of an overall assessment of the use of indicators in vulnerability assessment, their advantages and drawbacks as well as potential improvements of indicator-based methods.

1. Introduction

The plethora of tools for the assessment of risk, vulnerability, exposure and other aspects related to disaster management has also a downside. Practitioners are often confronted with the following question: what does the choice of different tools mean for the outcome and the final decision making?

Decision makers are mainly in need of two kinds of information: information regarding the hazardous process (expected magnitude and frequency, time of occurrence, spatial extend) and information regarding the elements at risk. As far as the elements at risk are concerned, an inventory of potentially affected buildings and infrastructure is of high significance. Moreover, the location of vulnerable people and the spatial pattern of social characteristics of these people such as age, disability, income and education level are important for decision making, since previous studies have shown that such characteristics influence the effectiveness of risk mitigation [1]. Presumably, not all elements at risk are equally susceptible to the impact of natural hazards, and as such their physical vulnerability is different [2]. As far as buildings are concerned, characteristics including material, dimensions, shape and surroundings play a major role to assess vulnerability [3,4]. While the use of indices is more common for the assessment of social vulnerability [5,6], the physical dimension of vulnerability has repeatedly been captured using mathematical functions describing a relationship between the proportional loss and the hazard magnitude. Different vulnerability curves are used for different types of hazards such as river floods [7], debris flows [8–10], earthquakes [11], and tsunamis [12]. In order to deduce these functions, however, empirical data are required. These data include the observed process magnitude and the corresponding loss ratio for each building. As reviews have shown, the resulting curves are often case-specific and can hardly be transferred to other regions [13], such as those vulnerability functions proving validity for torrential hazards in the European Alps [14] cannot be 1:1 transferred to torrential hazards in Mediterranean countries [15,16]. Consequently, risk mitigation and event management still remains a challenge and decision-makers and practitioners are often overwhelmed when actions to reduce losses have to be set on short-notice during an event [17].

To overcome this gap, indices, or composite indicators have repeatedly been used to support decision-making [13]. However, although they may facilitate risk mitigation, emergency management and communication through a quantitative interpretation of complex issues,

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they may be misinterpreted and, therefore, misleading [18]. Using the example of tsunami hazard, we present an approach to overcome this challenge and to propose a method to better acknowledge the uncertainties behind vulnerability assessment.

With respect to European countries, tsunami is a rare and underdocumented natural hazard in comparison to other natural processes such as earthquakes and floods [19]. Lack of empirical data on tsunami damage prior to 2004 led to the development of the PTVA (Papathoma Tsunami Vulnerability Assessment) model [20,21] which aimed at assessing the vulnerability of buildings at areas where no loss data were available. The PTVA focused on the development of a relative vulnerability index (RVI) for every building based on building characteristics that affect the capability of the building to withstand the hazard impact. As such, the lack in empirical loss data can be overcome, and different options in risk mitigation can be better developed and planned in collaboration between hazard experts and emergency managers and other decision makers.

The construction of indicators involves several steps including indicator selection, normalization, weighting and aggregation into a final index [22]. Indicator weighting is the most sensitive step for index construction, and can either be based on statistical assessments (such as Principal Component Analysis (PCA) or factor analysis) or on participatory approaches (such as the Analytic Hierarchy Process (AHP) or the Budget Allocation Process; [22]. The overall aim of indicator weighting is (a) to investigate any correlation among indicators in order to detect overlapping information and (b) to select a suitable weighting and aggregation approach for the final index calculation. While statistical approaches rely on a considerable amount of robust data, participatory approaches entail subjectivity, however, expert judgment is often based on many years of experience which may be actually a beneficial aspect [23].

Irrespectively from the method applied, a comparison of 106 studies for index construction in risk assessment revealed that the most common approach used (41.5%) was the “equal weights” method [24], where different indicators contribute to the final index with the same portion, such as the Human Development Index (HDI) [25], the Environmental Vulnerability index (EVI) [26] and the Social Vulnerability Index (SoVI) [5]. Some studies (13 studies) used weights that were defined by the authors themselves, such as the ESPON vulnerability index [27]. The remaining studies were either based on participatory methods (19 studies) or statistical methods (28 studies). The AHP indicator weighting was the most popular participatory approach used, such as in the Risk Management Index [28] which engaged local authorities for the pairwise ranking of the attributes. Obviously, the choice of weighting techniques directly affects the value of the index. The importance of weighting methods and their effect on the value of an index is highlighted through three case studies by Becker et al. [29]. Three indices (the *Resource Governance Index*, the *Good Country Index* and the *Water Retention Index*) were estimated using original and optimized weights showing a significant shift in the ranking. The optimization of the weights of the *Good Country Index*, for example, resulted in countries improving up to 86 places (out of 125 countries) [29]. The comparison of three multivariate social vulnerability methodologies applied on the same dataset also showed that different methodologies lead to significantly different results, however, the authors insist that additional research using different sets of indicators and scales is necessary [30].

The overall aim of the present research is to apply to the same case study area the two most recent versions of the PTVA method (PTVA-3 and PTVA-4) in order to demonstrate the importance of weighting vulnerability indicators for the development of vulnerability indices and its implication to decision making. The results show that different weighting leads to a different spatial vulnerability pattern which, eventually, leads to different decisions taken by authorities, planners and emergency services.

Dall'Osso [31], after introducing PTVA-4, applied the two versions at the same coastal segment and compared the results. This coastal segment, however, has very different characteristics in comparison with the present case study area. The methods were applied on 2211 buildings in Botany Bay in Sydney (Australia), a very densely-populated and diverse area as far as land use is concerned. The area is also very important since significant infrastructure such as Sydney's International Airport and main commercial harbour are also located there. The comparison of the results mainly focused on the technical differences between the methods rather than their implications for decision making. However, in the present paper the methods are applied locally in small and relatively homogenous coastal segments.

The most recent study applying and validating the PTVA-3 and PTVA-4 method has been carried out in two coastal cities in Chile [32]. The study validates the method by using damage data from a relatively recent event (16 September 2016).

The paper is organized as follows. First, the methodological approach is described and the two models and their differences are explained. The section is organized in sub-sections describing the tsunami hazard in the area, the worse-case scenario, the field work and the results. Finally, the results are discussed in section 4. The discussion expands to the effectiveness of index-oriented methods and their implication in decision making in the field of disaster management and disaster risk reduction.

2. Methodological approach: PTVA-3 and PTVA-4

Following the 2004 and 2011 tsunamis many fragility and vulnerability curves were suggested [12] but since they were tailored to events in southeast Asia and Japan they could not be transferred to other tsunami-prone areas with different architecture and building design. The Papathoma Tsunami Vulnerability Assessment (PTVA) method provided an alternative approach based on vulnerability indicators rather than empirical loss data.

The PTVA method makes use of vulnerability indicators (i.e. characteristics of the buildings regarding their properties and surroundings) to qualitatively assess risk and provide decision-support in hazard management. The model has been revised and updated four times (PTVA, PTVA-2, PTVA-3 and PTVA-4) and has been also validated following the tsunami of 2004 [33]. Moreover, the PTVA method has been applied by several scholars in different case studies including Italy [34–36], Portugal [37], Morocco [38,39], Egypt [40], Japan [41], Iran [42] and Chile [32].

The first version of the method was quite trivial. It considered the basic characteristics of the buildings such as the building material, the number of floors and the surroundings. The weights of the indicators were assigned by the authors based on past records of events and their aggregation was based on a weighted linear combination. Later on it was modified to include information regarding the height of the tsunami inundation (PTVA-3). The indicators used for the construction of the vulnerability index were weighted using M-Macbeth, a software used for multi-criteria analysis [43]. In the latest version (PTVA-4), the indicators were weighted in a slightly different way based on a questionnaire distributed to a large number of tsunami experts worldwide.

In the present paper the two latest versions of the PTVA method were applied, and their results compared with respect to the application in risk management. In more detail, the flow chart of the methodological steps is demonstrated in Fig. 1.

As a first step, the inundation extent and magnitude of a real-case scenario is determined. For each building located within the potentially inundated area, vulnerability indicators are collected. Ideally, the data may be available by the local authorities, but in most cases, due to the high resolution and detail of the required information, data collection is carried out through extensive field work. The collected data are entered in a geodatabase, and the RVI is calculated based on equation (1):

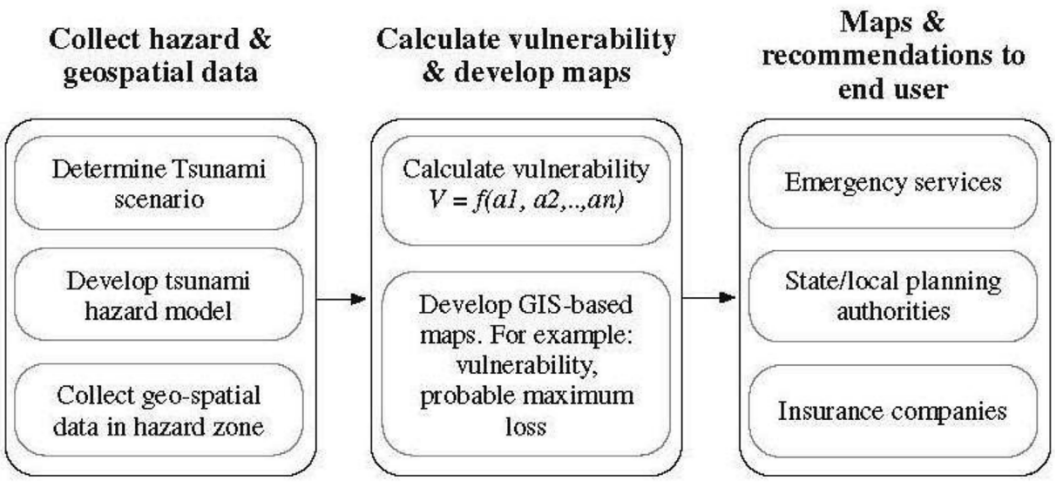


Fig. 1. The methodological steps of the PTVA model [43].

$$RVI = \frac{2}{3}(SV) + \frac{1}{3}(WV) \quad (1)$$

[44].

The index has two components: the structural vulnerability (*SV*), which depends on the building vulnerability (*Bv*), the exposure (*Ex*) and the level of protection (*Prot*), (equation (2)) and the Water Vulnerability (*WV = 0, 1*) which is related to the extent to which a building is submerged by water (number of inundated levels)/(number of total levels) (equation (3)).

$$SV = (Bv)\ \text{x}\ (Ex)\ \text{x}\ (Prot) \quad (2)$$

$$WV\ (0,+1) = \text{(number of inundated levels)/(number of total levels)} \quad (3)$$

Every building considered is given a value for each of the indicators related to *Bv, Ex* and *Prot,* such as building material, number of stories, surrounding wall, etc. (Tables 1 and 2). The values of the *Bv*, *Ex* and *Prot* variables range from −1 to +1 according to whether the specific characteristic contributes positively or negatively to the overall vulnerability of the building. The component *Ex* represents the exposure which is calculated on the basis of the water depth impacting the building.

The *BV* and *Prot* variables are obtained through the following equations (4) and (5):

$$Bv\ (-1,+1) = w1 \cdot \cdot s + w2 \cdot m + w3 \cdot g + w4 \cdot f + w5 \cdot so + w6 \cdot mo + w7 \cdot pc \quad (4)$$

and

$$Prot\ (0, +1) = w1 \cdot Prot\ br + w2 \cdot Prot\ sw + w3 \cdot Prot\ nb + w4 \cdot Prot\ w \quad (5)$$

where w1, w2 etc. are the weights shown in Tables 1 and 2

The RVI is calculated for each building using the AHP weighting approach based on personal judgement (PTVA-3) and the AHP based on expert judgement (PTVA-4) each one leading to slightly different results. The difference in weighting between the PTVA-3 and the PTVA-4 model is demonstrated in Table 1 (for *Bv*) and in Table 2 (for *Prot*).

All the components of the *RVI (SV, BV, Ex, Prot* and *WV*) have an original and a scaled score for each building (Table 3).

In both PTVA-3 and PTVA-4 the chosen weighting method is the AHP, a participatory weighting method based on a pair-wise comparison of the indicators which captures not only the preference between two different indicators but also the strength of preference [18]. The method was chosen to improve the weighting of the original model which was arbitrarily determined by the authors themselves. Subjectivity remains an issue since AHP is a participatory method, however, the software used may detect inconsistencies in the judgements of the users and remove them.

The PTVA-3 model uses AHP to calculate weights based on the authors opinion and judgement prior to the Tohoku tsunami in 2011. In the PTVA-4 model the indicators are re-weighted considering information acquired following the Tohoku tsunami. The weighting involved a questionnaire which was filled by 14 authors of scientific papers in the field published between 2005 and 2015. In this questionnaire scholars were asked to suggest weights for a list of indicators regarding the structural vulnerability of buildings to tsunamis and the protection from the tsunami impact. Additionally, experts could provide (and weight) new indicators as well as make open comments [31]. The difference in the weighting of PTVA-3 and PTVA-4 can be seen in Tables 1 and 2. The two models have additional differences concerning the re-scaling method, the calculation of the *WV* and the classification method of the RVI. However, in this paper, we keep the same re-scaling method, *WV* calculation and classification method for both models to determine the consequence of the different weighting in the results.

The spatial pattern of the RVI for both approaches was then visualized through vulnerability maps of the area. These maps may be

Table 1
The weights of the PTVA-3 and the PTVA-4 models for each indicator determining the *Bv* [31].

Attributes of the building	PTVA-3 weight (0–100)	PTVA-4 weight (0–100)
number of stories (*s*)	100	85
main construction material (*m*)	80	100
ground floor hydrodynamics (*g*)	63	69
foundation type (*f*)	60	69
shape and orientation (*so*)	46	52
preservation condition (*pc*)	23	34

Table 2
The weights of the PTVA-3 and the PTVA-4 models for each indicator determining the *Prot* [31].

Attributes of the building	PTVA-3 weight (0–100)	PTVA-4 weight (0–100)
building row (*br*)	100	100
seawall (*sw*)	73	84
natural barriers (*nb*)	73	72
large movable objects (*mo*)	51	58
brick wall around the building (*w*)	55	42

Table 3
Variables used for the calculation of the RVI scores [44].

RVI	1–1.8	1.8–2.6	2.6–3.4	3.4–4.2	4.2–5
Description of relative vulnerability level	MINOR	MODERATE	AVERAGE	HIGH	VERY HIGH
Relative Vulnerability index (*RVI*) = (2/3) x (*SV*) + (1/3) x *WV* (Eq. (1))					
SV (original)	1–25	25–50	50–75	75–100	100–125
SV (scaled)	1	2	3	4	5
SV = (*BV*) x (*Ex*) x (*Prot*) (Eq. (2))					
Bv (original)	−1 to −0.6	−0.6 to −0.2	−0.2 to +0.2	+0.2 to +0.6	+0.6 to +1
Bv (scaled)	1	2	3	4	5
Ex (original)	0–1 m	1–2 m	2–3 m	3–4 m	> 4 m
Ex scaled	1	2	3	4	5
Prot (original)	0 to 0.2	0.2 to 0.4	0.4 to 0.6	0.6 to 0.8	0.8 to 1
Prot (scaled)	1	2	3	4	5
WV = (number of inundated levels)/(total number of levels) (Eq (3))					
WV (original)	0 to 0.2	0.2 to 0.4	0.4 to 0.6	0.6 to 0.8	0.8 to 1
WV (scaled)	1	2	3	4	5

interpreted and used for decision making by the end-users. In the following chapters the application of PTVA-3 and PTVA-4 is presented.

3. Application to the case study of Sant'Isidoro and Torre Squillace (Apulia, Italy)

3.1. Tsunami hazard in the Mediterranean with a focus on Italy

The collision between the Eurasian and African plates is mainly responsible for the seismicity and volcanic activity of the Mediterranean area. Many Mediterranean coastal areas such as the southern part of Spain, the Tyrrhenian basin, the subduction of the Calabrian arc, the Hellenic arc and the Cyprean arc have been often affected by tsunamis in the past [45]. There are several catalogues that list tsunamis that have occurred in the Mediterranean but also catalogues of individual countries such as Greece or Italy [46]. Maramai et al. [47] have compiled a catalogue for tsunamis in the Mediterranean which lists 290 events since 6150 BC. The majority of Mediterranean tsunamis have been triggered by submarine earthquakes and less frequently by inland earthquakes. Some tsunamis in the area were generated as a consequence of a terrestrial or a submarine landslide. Less than 5% of the tsunami were originated by volcanic activity, mainly caused by Vesuvius and the Aeolian Islands volcanoes in Italy [47]. Papadopoulos et al. [48] suggest that the mean frequency of strong tsunamis in the Mediterranean is 1 every 96 years.

As far as Italy is concerned, the newest tsunami catalogue demonstrates that since 79 AD at least 67 events have been recorded. However, the first 1000 years of the catalogue are not well recorded and the reliability of the recorded events varies [46]. In order to support maritime policy and Integrated Coastal Zone Management (ICZM) in Italy, a geodatabase was created containing all the scientific contributions concerning tsunamis that occurred along the Italian coastline. The database is available online and contains papers published after 1990 [49]. Despite the large number of events recorded, in-depth research and vulnerability assessments in Italy are concentrated in areas with a record of especially strong and destructive tsunamis, such as Eastern Sicily and Calabria. A vulnerability assessment study was conducted in Syracuse, Sicily using the PTVA-3 and the SCHEMA method. Although the study points out the advantages and disadvantages of the two methods, it concludes that the results of the two approaches can be put in parallel although in some cases SCHEMA may result to lower level of expected damage [36].

3.2. Tsunami hazard in Apulia: description of the study area

In Apulia, where the study area is located, four events are recorded in 1627, 1731, 1743 and 1889. However, given the incompleteness of the catalogues and the tsunamigenic areas in the vicinity of Apulia, this does not mean that the area has a low tsunami hazard. On the contrary, large boulders found on the inner (facing Calabria) [50] and the outer coasts (facing Albania) of Apulia show that tsunami activity may be the reason for their transport inland [51]. Mastronuzzi et al. [51] investigated the presence of 80 t large boulders up to that were scattered in tracts of the gently sloping rocky coast along the Ionian coast of Apulia. A detailed study was carried out using boulders found in the small town of Torre Squillace, a suburb of the more known Porto Cesareo and part of the province of Lecce (Fig. 2). The authors concluded that given the origin of the boulders the only explanation for their appearance at the coast is their transportation inland by a tsunami. In more detail, Mastronuzzi et al. [51] suggest that the boulders were transported during one of the following events: the tsunami of April 1667 which was also responsible for the destruction of the city of Dubrovnik (Croatia) or the tsunami associated with the earthquake that struck the region in February 1743. Later on and based on this evidence, Pignatelli et al. [52] calculated the maximum extend of flooding in the area considering the initial position of the boulders, their size and weight and the rock density. They produced a flooding map of the area and they calculated the possible tsunami height in different locations. At Torre Squillace, which is also one of the two case studies of the present paper, Pignatelli et al. [52] concluded that the minimum tsunami height must have been 4.94 m and the inundation limit approx. 130–139 m inland. In the present paper two coastal segments are investigated: Torre Squillace and Sant'Isidoro, 4 km further south, a neighbouring settlement which was chosen due to its building and land-use diversity in comparison with Torre Squillace.

In more detail, the study area is located in the Salento peninsula, at the southern emerged part of the Adriatic Plate which constitutes the foreland of the Apennine mountains [53]. The Salento Peninsula comprises a Variscan basement covered by a 3–5 km deep Mesozoic carbonate layer. The region is covered by thin deposits of Tertiary and Quaternary age [54]. The landscape is rather flat and composed of different Pleistocene plains. These plains are situated at several levels above the sea level until an elevation of 160 m. They border with degraded fault scarps and by relict cliffs [53]. The coastal landscape is characterized by marine terraces that were produced by superimposition of the regional uplift and glacial sea level changes that have continued since the Middle Pleistocene period. The terraces are usually arranged in a staircase morphology. The coastline is characterized by beaches, gently sloping coasts, convex rocky coasts and cliffs [54] (Fig. 2). The beaches are generally bordered by a dune belt and sometimes backed by swamps. Convex rocky coasts are, usually, shaped on limestone. The cliffs along the coast are composed either of clays or calcareous sandstones or highly karstified limestone [54].

Torre Squillace is a settled coastal area that belongs

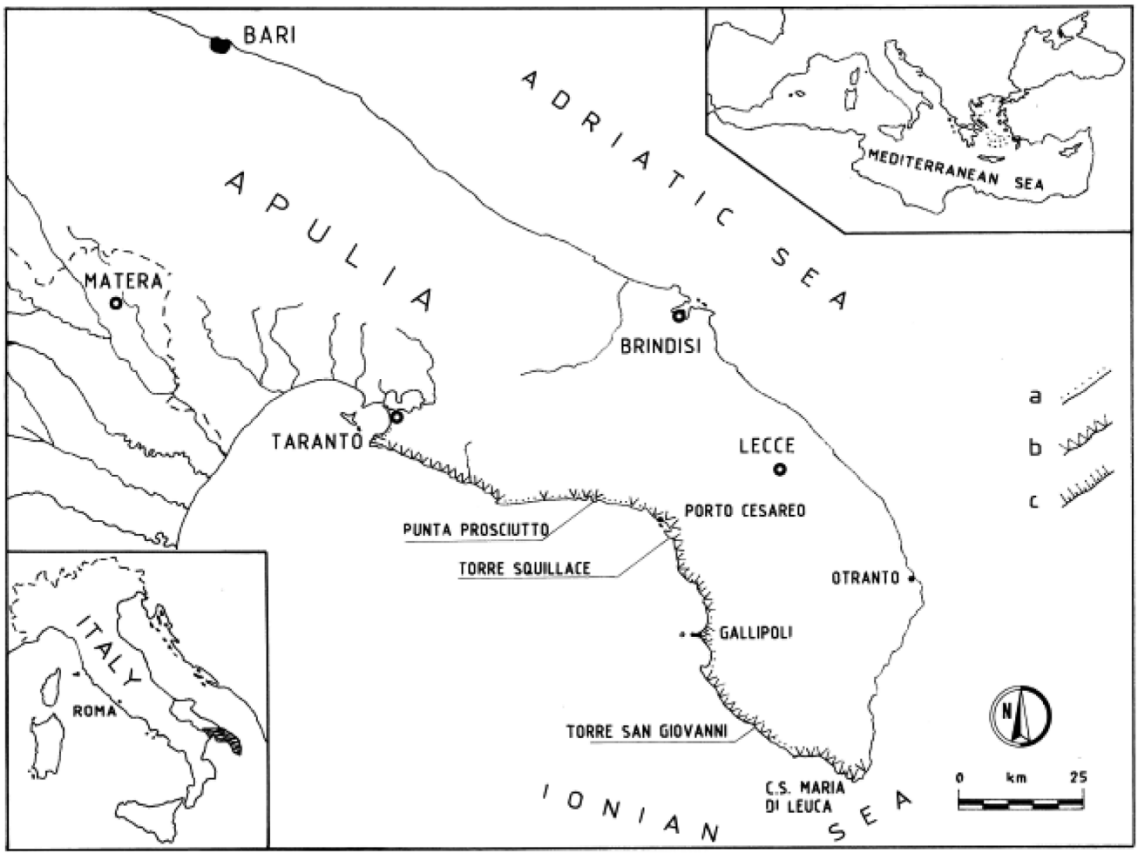


Fig. 2. Torre Squillace and other areas where the existence of large boulders was investigated. (a: beaches, b: gently sloping and convex rocky coasts, c: cliffs) [51].

administratively to the Nardo borough and has 144 permanent residents. There are 849 buildings in the area of which 742 are residential. Sant'Isidoro is also part of the municipality of Nardo. It has 186 permanent residents and 870 buildings (806 residential). In both areas, most of the buildings have only one floor and are mainly used in the summer by non-permanent residents.

3.3. Application of PTVA-3 and PTVA-4 and results

The first step was to define the worst case scenario which will also guide the delimitation of the tsunami inundation zone. Based on the results of Mastronuzzi et al. [51] and Pignatelli et al. [52], we considered the event of 1743 as the worst scenario for the study area. The potential inundation area according to this scenario was identified to be the area between the shoreline and the 5 m contour, since the tsunami wave of 1743 had a maximum height of 5 m [52]. On the basis of heterogeneity, a representative sample of buildings was chosen in both case study areas to apply PTVA-3 and PTVA-4. In more detail, the methods were applied at 193 buildings all located between the shoreline and the 5 m contour. Fig. 3 offers a graphical presentation of the inundation zone. In Fig. 4 some characteristic buildings of the area, as well as the beaches of Sant'Isidoro and Torre Squillace, are shown. Most of the buildings are one-floor brick buildings mainly used as summer residences.

The two methods were applied for each sample building in both areas. Information regarding the indicators presented in Tables 1 and 2 were collected and an RVI was calculated for each building. The results for Torre Squillace are shown in Table 4 and the spatial pattern of the RVI index is demonstrated in Fig. 5 for both PTVA-3 and PTVA-4. Similarly, PTVA-3 and PTVA-4 was also applied in Sant'Isidoro and the results can be seen in Table 4 and Fig. 6, respectively. In Torre Squillace for the PTVA-4 the vulnerability class with the highest building number is the average class, whereas for the PTVA-3 the class with the most buildings is the very high class. The percentage of buildings belonging to the very high class for PTVA-4 are only around 50% of the percentage of buildings belonging to the same category for the PTVA-3. Generally speaking, one of the first observations is that PTVA-4 is more conservative in comparison with PTVA-3 in the sense that PTVA-3 tends to characterize more buildings as highly vulnerable whereas during the application of PTVA-4, vulnerability levels were decreased, characterizing a very large number of the buildings as buildings with "average" and "moderate" vulnerability.

As far as the spatial distribution of the RVI is concerned, the implications that the different results can have on planning and decision making are quite obvious. In Fig. 5, it is clear that by the application of PTVA-3 the whole first row of buildings facing the coast are characterized as very highly vulnerable. On the other hand, the application of PTVA-4 shows that only sporadic buildings belong to the very high vulnerability category. In the first case, local authorities and other decision makers could use the map to support the construction of structural protection measures that would protect the vulnerable "façade" of the settlement, whereas the second map could direct authorities towards the reinforcement of individual buildings or the development of local adaptation measures for individual buildings.

As far as Sant'Isidoro is concerned, PTVA-4 is again the more conservative method of the two, characterizing less buildings as very highly vulnerable in comparison with the PTVA-3 (Table 5). It is also remarkable that although PTVA-3 classifies only 4.2% of the buildings as moderately vulnerable, for thePTVA-4 the percentage of the same class is 22.5%. In this area there are no buildings in the minor class. In Sant'Isidoro, PTVA-3 classifies a very large percentage of the buildings (76.1%) in the very high and high vulnerability category. In contrast, buildings in the very high and high category for the PTVA-4 comprise 52.15% of the total number of buildings. In this case, local authorities

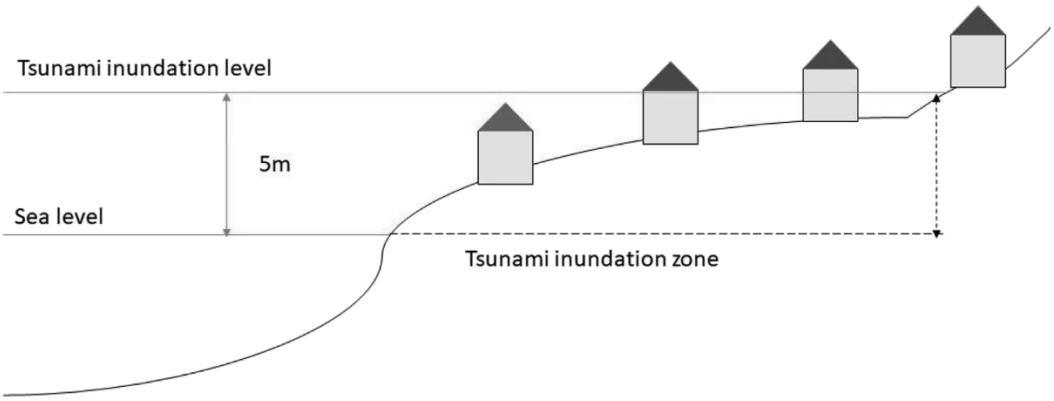


Fig. 3. A graphical representation of the chosen inundation zone.

would have a stronger argument to require funds from the central government using the results of PTVA-3 rather than the results of PTVA-4.

The spatial distribution of the RVI in Fig. 5 shows that although PTVA-3 indicates that most of the highly vulnerable buildings are mainly located at the coastline, it also categorizes a number of buildings as very highly vulnerable although they are located at a considerable distance from the shoreline, being shielded by other buildings. Moreover, some buildings being classified as highly vulnerable by PTVA-3 drop 2 categories and are classified as moderate by PTVA-4. As far as the implications for decision makers are concerned, PTVA-4 suggests that the very highly vulnerable buildings are the ones facing the shore guiding decision makers to focus their efforts in this area. Finally, in some cases, the two methods point at completely different buildings characterizing them as highly vulnerable. Such a case we can see at the southwest side of the map in Fig. 6. The highly vulnerable building in the series of detached houses is not the same for PTVA-3 and for PTVA-4 leading the decision makers (that in this case may be the house-owners) to different decisions.

Table 4
Distribution of the RVI results for Torre Squillace (%).

	Very high	High	Average	Moderate	Minor
PTVA-3	29.2	23.3	27.5	15.8	4.2
PTVA-4	15	11	38	30	6

The distribution of the results for both methods is similar to the ones of Izchierdo et al. [32] who also applied both models to a coastal area in Chile. The concentration of buildings in the average and moderate category for PTVA-4 in contrast with the concentration of buildings in the very high, high and average category for PTVA-3 for both case study areas can be explained by the difference in the weighting. In the case of PTVA 3 the most important indicator is the number of stories whereas for the PTVA-4 the indicator with the highest weight is the construction material. This difference consequently leads to different results.



Fig. 4. Characteristic buildings in Torre Squillace (top left) and Sant'Isidoro (top right) as well as a view of the beaches in Torre Squillace (bottom right) and Sant'Isidoro (bottom left).

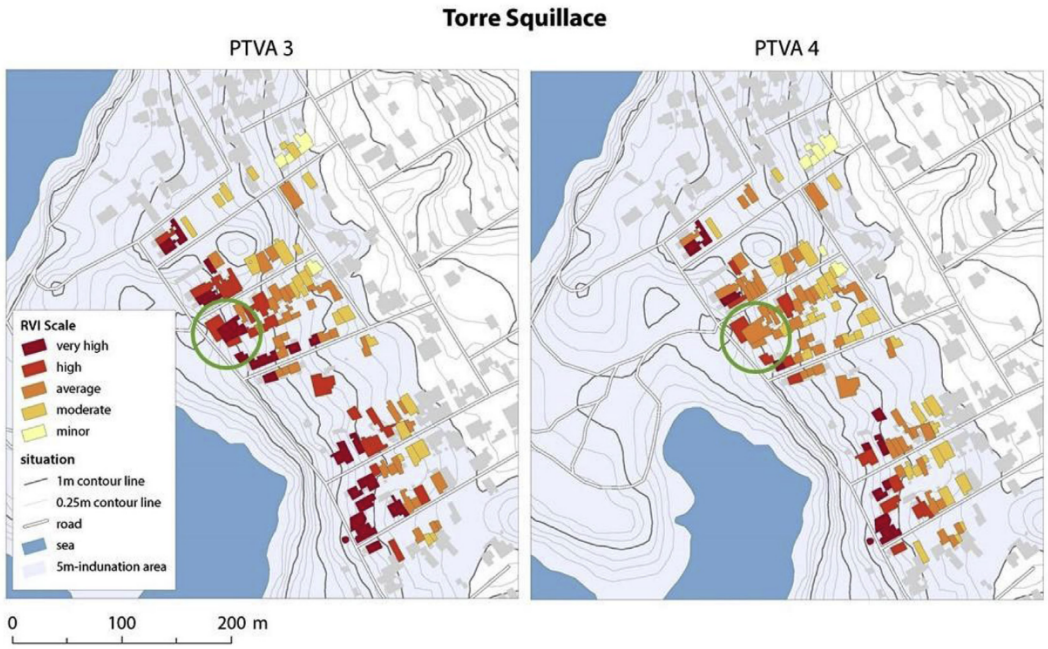


Fig. 5. The spatial pattern of the RVI in Torre Squillace: the green circle indicates commercial buildings that have been assigned to different vulnerability categories by PTVA-3 and 4. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

4. Discussion

The comparison of the results from the Italian case studies shows that PTVA-4 is more conservative classifying more buildings in lower vulnerability categories. This was also suggested by the results of the comparison study in Chile [32]. Izquierdo et al. [32] reconstructed the event and created an inundation map using field data. The buildings in the Chilean case studies demonstrate a greater variability than the ones in Italy in terms of building material and number of floors. Nevertheless, the results of the comparison between the two models were similar to the ones presented herein. The comparison of the PTVA-3 and 4 results with the actual damages of the 16 September 2016 tsunami showed that the results of PTVA-3 were closer to the observed damages.

Table 5
Distribution of the RVI results for Sant'Isidoro (%).

	Very high	High	Average	Moderate	Minor
PTVA-3	42.3	33.8	19.7	4.2	0
PTVA-4	33.8	18.3	25.4	22.5	0

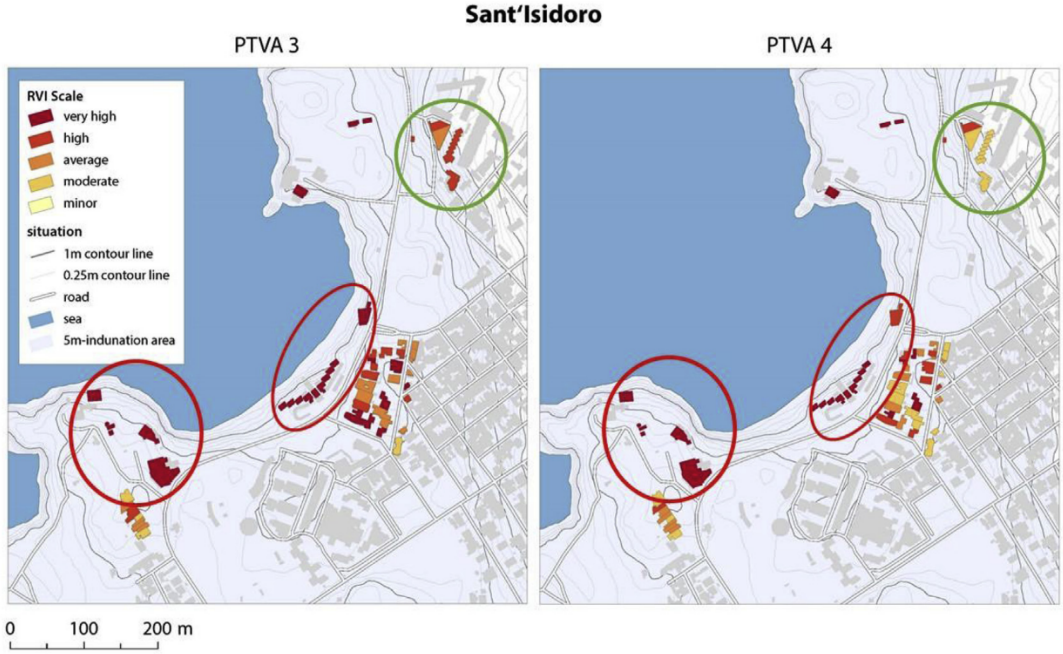


Fig. 6. The spatial pattern of the RVI in Sant'Isidoro: buildings in the red circles are assigned to the high vulnerability category by both versions of the model, whereas the RVIs of the buildings located within the green circles are different for each version of the model. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

This underestimation of vulnerability may lead to lower investments for retrofitting, awareness campaigns etc. that will reduce the level of preparedness of the community. At a household level, homeowners living in such low-vulnerability buildings may have an illusion of safety (safe development paradox) and do not undertake actions for self-protection [55]. Furthermore, insurance companies may underestimate possible consequences and lower their premiums, finding themselves in a difficult position should a tsunami strike and the vulnerability of buildings proved to be lower than assessed. On the other hand, a conservative vulnerability assessment leads to conservative investments that may be easier accepted by the central government and the local community for a hazard type associated with very low risk perception. In the following paragraphs the use of PTVA for different decision-making levels is discussed.

Decision making in disaster risk reduction is a process aiming at the improvement of the prediction and prevention capacities in the pre-disaster phase, the emergency response during and shortly after an event based on a sufficient resources allocation and at the organization of the reconstruction and the return of the affected community to normality. Decision making takes place in different levels: individual/household, business, community and governmental level.

4.1. Decision making for individuals: public awareness and participation

The resulting vulnerability maps, if publicly available, may raise awareness in the potentially affected community. Moreover, knowledge about own vulnerability levels may raise the chances of homeowners to invest in retrofitting or in local adaptation measures (e.g. small protective wall or vegetation) and vertical evacuation planning [56–58]. Moreover, vulnerability assessment results constitute a valuable tool to support the “build back better” during the reconstruction phase. A lower RVI following the reconstruction shows that through small changes of the building properties (e.g. surroundings, better preservation condition) vulnerability could be reduced or resilience could be increased. Maps such as those presented herein may also be used to support people-centred housing recovery, resulting in an increase in ownership and responsibility [59].

Table 6 demonstrates the distribution of households within the different vulnerability categories. In order to enable better decision-making for individuals, residents of buildings belonging to the very high and high vulnerability category should be informed so that they can develop options regarding reliable and cost-efficient adaptation measures (protecting wall or fence, location of sensitive products, possibilities of vertical evacuation). As shown in Table 6, it is evident that the number of households located in highly vulnerable buildings according to PTVA-3 is significantly larger than the number of households indicated by PTVA-4. In other words, the number of homeowners alerted by the results of PTVA-3 is nearly double than the number of individuals alerted by the results of PTVA-4.

4.2. Decision making for businesses

The buildings used in the present study are in the vast majority residential (Table 7), however, also few commercial buildings such as restaurants or shops and few hotels are exposed.

Table 6
Distribution of households in the vulnerability categories.

		very high	high	average	moderate	minor
PTVA-3	Torre Squillace	34	29	35	23	4
	Sant’Isidoro	13	31	23	8	0
	TOTAL	47	60	58	31	4
PTVA-4	Torre Squillace	14	12	50	39	10
	Sant’Isidoro	5	16	23	26	0
	TOTAL	19	28	73	65	10

Table 7
Building use in the study area.

	Residential	Commercial	Hotels	TOTAL
Torre Squillace	119	2	1	122
Sant’Isidoro	44	26	1	71

Businesses operating (hotels, restaurants) or having interests in an area (e.g. insurance companies) may use the resulting vulnerability maps to adopt local adaptation measures to protect their guests from a possible tsunami attack. Similar results were also described by Karagiorgos et al. [60] with respect to micro-sized enterprises exposed to flash flood hazards in coastal areas of Greece. Such local adaptation may include removal of movable objects from the ground floor and reconsideration of the use of the ground or first floor(s) enabling the way to the terrace to facilitate vertical evacuation. A collection of actions that hotel owners may take to be prepared for a tsunami attack can be found in the “Guide to Tsunami for Hotels” [61] prepared by the North-eastern Atlantic and Mediterranean Tsunami Information Centre.

In more detail, in Fig. 5, the green circle shows a group of buildings with commercial use that have been assigned to the high vulnerability category by the PTVA-3 and to lower vulnerability categories by the PTVA-4. Apart from general information on risk mitigation, the temporal dynamics of vulnerability can be of particular interest if commercial buildings are considered. Firstly, buildings used as hotels and pensions may host a large number of people during some periods of the year (e.g. summer), which is a challenge for evacuation planning, as shown in Fig. 6 (red circle) for the village of Sant’Isidoro. Moreover, these tourists often lack local knowledge and may have very low tsunami awareness. Secondly, owners of commercial buildings have to consider a loss of products in case of a tsunami event and, therefore, have to be informed regarding local adaptation measures or alternative storage opportunities. In case of larger businesses operating within the inundation zone the impact on the local economy has to be considered and structural measures for the protection of this kind of buildings may have to be taken. Thirdly, vulnerability information may be also used by insurance companies as a guide for setting insurance premiums. Based on the RVI and individual indicators, recommendations may be made by insurance companies to the building owners to decrease their vulnerability by making some small changes offering reduced premiums in exchange.

4.3. Decision making for authorities and emergency services

Decision making is relevant to all the phases of the disaster cycle, as shown in Fig. 7 and Table 8 [62,63], and may therefore be enhanced during prevention, preparedness, response and recovery. However, decision making takes often place under time pressure, with limited resources and data, as well as under a number of uncertainties mainly deriving from lack of information. Contemporary technology may be a valuable tool in the hands of decision makers (GIS, Mobile Apps for data collection, social media etc.), nevertheless, technology can increase the level of complexity [64].

A survey among professionals working in the emergency management community showed that the most important factor influencing decision making is access to reliable knowledge coming from experts [65]. In the same survey, decision makers in emergency management also pointed out their main challenges including data quality and data availability especially for low frequency-high consequence events.

The research presented herein clearly shows that the PTVA method may constitute a valuable tool for decision-making. In Table 8, the relevance of the method for each decision group is shown. The database generated, as well as the results of the PTVA, may be used before the event (*Meta decision*) to organize public awareness or the structural protection in different scales (municipal or individual level) of

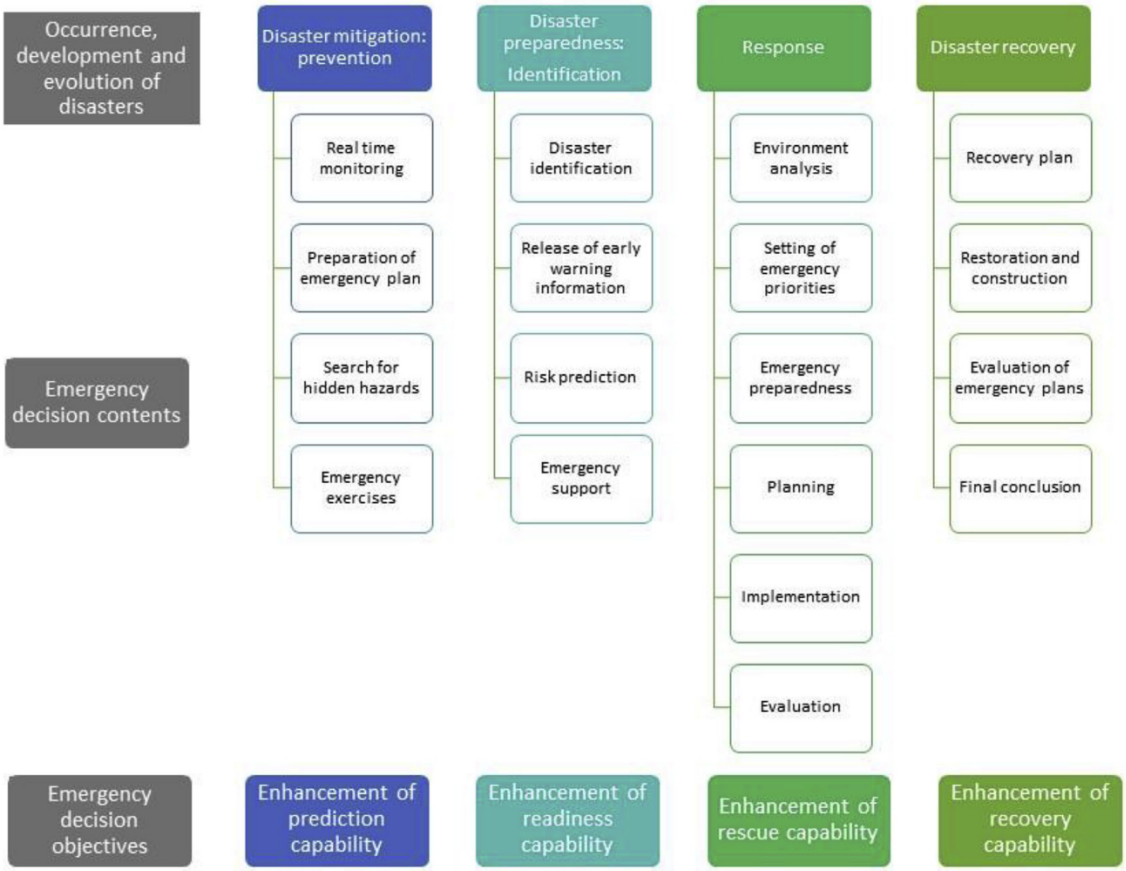


Fig. 7. The process of emergency decision making in the field of natural hazard management (modified from Zhou et al. [62]).

vulnerable buildings. Vulnerability maps like the ones presented in Figs. 5 and 6 may be presented to the public. The owners and inhabitants of the vulnerable buildings may be invited to information evenings focusing on emergency response possibilities during a tsunami event. The maps may help raising awareness in the area and change the risk perception which may be currently low. In the present study no assessment of the risk perception of the population was carried out. However, Cerase et al. [66], carried out more than 1000 interviews in two tsunami prone regions in Calabria and Apulia to investigate tsunami awareness and risk perception. Although the study is not completed to date, the first results show that residents of these areas appear less misinformed on tsunamis as expected, nevertheless, the risk of tsunamis is underrated or neglected in these areas. It is also worth mentioning that the residents have high expectations of the authorities as far as risk management is concerned. Similar to the European Floods Directive [67], the vulnerability map can be used as a basis for an emergency plan utilized by rescue services for the identification of buildings with high concentration of victims during the response phase (*Operational decision*). PTVA results could support tools for the identification of potential tsunami vertical evacuation sites, that, until now are often solely based on an inundation scenario [68]. Finally, in the reconstruction phase (*Planning decision*), the method itself may be validated and the list of indicators as well as their weighting may be reconsidered and updated. The comparison of the results before and after may also highlight a possible "build back better" or "build back worse" if this is the case, emphasizing the need for resilient reconstruction strategies [69]. Finally, vulnerability maps can enhance risk communication in the local community and raise the risk perception in the area.

It is highly possible that fatalities may be located in highly vulnerable buildings (Table 6), therefore, in our case study areas, emergency services should direct their efforts in these buildings following the occurrence of a tsunami. In Figs. 5 and 6 it is clear that a significant amount of buildings is characterized by both versions of the model as

Table 8
The relevance of the PTVA for each type of decisions (inspired by Platt [63].

Decision group	Decision makers	Relevance of the PTVA
Meta decisions (before the disaster strikes)	Politicians, policy makers	Construction of structural protection measures designed to protect specific buildings (indicated by the PTVA) Focusing on local communities (living in buildings indicated by the PTVA) to raise awareness and to implement training courses for response. PTVA can also guide decisions related to the allocation of funds
Operational decisions (as a response to a crisis)	Disaster managers	Focus the response, relief and early recovery in specific parts of the town soon after a disaster strikes based on maps such as the ones presented in Figs. 4 and 5. PTVA can also be used as a basis for an evacuation plan.
Planning decisions (during the recovery and reconstruction phase)	Policy makers, urban planners, economists	PTVA can be used after the reconstruction phase has been completed to compare the post-event to the pre-event vulnerability and to monitor the "build back better".

highly vulnerable, however, this is not always the case. In Fig. 6, the area in the green circle may be a priority for the emergency services if they use PTVA-3 but not if they use PTVA-4. Nevertheless, buildings located in the red circles are a priority for the emergency services for both versions of the model. Moreover, emergency services may use the spatial distribution of single indicators (such as the number of floors) to locate buildings that cannot offer vertical evacuation possibilities. Moreover, residents with mobility difficulties and other vulnerable groups (elderly, children, patients) have to be identified so that evacuation and event management can get more efficient.

Additionally, vulnerability maps may be displayed together with additional information to highlight local challenges as well as solutions. The comparison of the RVI map with data concerning the income of the owners may reveal why some houses are of poor quality in comparison with others. The use of highly vulnerable buildings is also indicative. If the highly vulnerable buildings house public institutions, such as schools, old people's homes or kindergartens, then, their use and license to operate has to be reconsidered and potential relocation may be recommended. Additionally, the PTVA results may be displayed together with population data. PTVA-3 has been already applied in combination with the building population estimation model in Japan showing to which extend the population may be affected by a tsunami event [41]. The more conservative PTVA-4 in this case would give a completely different picture of the potential threat to the population.

Last but not least, the maps may support risk communication at a local level, as does the currently implemented European Flood Directive for flood hazards [70]. The PTVA model may be included in decision support tools or knowledge explain tools as it has been done in the past for other types of hazards such as flood hazards [71].

4.4. Reflection on the method

The application of the two versions of the PTVA highlighted general advantages of using indices for the assessment of physical vulnerability. First of all, the development of detailed inventories of elements at risk is encouraged. The information contained in the inventory databases may be further used for the improvement of building indices, however it can also be used individually (e.g. map showing the building condition) and can be updated easily. Moreover, vulnerability indices may be calculated for areas without an historic record of events, and as such, no empirical data is needed. Nevertheless, an indicator-based model cannot predict losses or translate vulnerability into monetary terms. It may, however, indicate where considerable losses may be expected. The information required for the development of the indices can be collected by non-experts. Furthermore, the visualization of the spatial pattern of indices is a good basis for planning (land use and emergency planning). These common databases can also act as a platform for multi-stakeholder collaboration and improvement of risk communication. Last but not least, the results of the method may also be used to raise public awareness in the area increasing in this way the participation of the local community in decision making. Awareness may contribute to a better response during and shortly after a potential event but it may also support low-cost local adaptation measures (e.g. surrounding vegetation, use of the first floor, access to terraces where people can find immediate shelter following an earthquake).

Index-oriented methods for vulnerability assessment have also drawbacks because they highly depend on data availability. A large amount of detailed information for each individual building is required which in most cases is not available and, therefore, has to be acquired through time-consuming field surveys. Moreover, the method is qualitative classifying buildings in vulnerability categories. This fact makes the method less attractive for practitioners that prefer the use of quantitative methods (e.g. vulnerability curves) to gain an overview on possible monetary loss, despite the uncertainties associated with them. As far as the PTVA method is concerned, the main disadvantage is that through scaling and normalization of scores a significant amount of information is lost. Considering the significant differences in the results deriving from PTVA-3 and PTVA-4, further studies are suggested in order to develop a reliable weighting system through the collection of additional data and the correlation with damage costs. This correlation could give better insights in the hierarchy and importance of the different building characteristics.

5. Conclusions

Two versions of the same method each one using a different weighting technique for indicators expressing tsunami vulnerability were applied to the same case study. The application showed that different weighting of the indicators leads to different results which may guide decision makers in diverse conclusions and actions. The implications in decision making for prevention, response and reconstruction were investigated in three decision making levels: individual/household, business and authorities/emergency services. Although the method has a number of advantages that in most cases outweigh the disadvantages, additional research has to be carried out in order to acquire a reliable tool for decision makers that can be used in all the different phases of the disaster cycle.

Through the application of the PTVA-3 and 4, the following recommendations for the improvement of the method are given:

- Additional research has to be undertaken to improve the weighting technique. The application of two different weighting methods showed that the results may lead to different decisions.
- Alternative data collection methods may be employed including remote sensing and questionnaires to achieve an up-to-date high-resolution database of the elements at risk.
- Validation of the method may be carried out following future events or by using data from well-documented past events.
- The method could be transferred to other places in the Mediterranean that have experienced tsunamis, such as Kos island in Greece which experienced a small but well-documented tsunami following an earthquake in the summer of 2017.
- The method has been applied in a potential inundation zone based on a study investigating the origin of large boulders at the coast. Modelling taking into consideration potential earthquakes generated in existing fault zones, the bathymetry of the littoral zone as well as the topography of the area may result in a more reliable tsunami run-up area as a basis for the vulnerability assessment.
- Finally, as also shown by Tavares et al. (2017), future research should focus on the combination of physical vulnerability models with population data, which includes not only population density but also socio-economic data (e.g. age, disability, gender etc.).

So far, the PTVA model remains a valuable tool to assess vulnerability to tsunamis. Although the PTVA model cannot predict loss, it can give an overview of buildings that need to be retrofitted, and can be used in decision making by offering evacuation opportunities and supporting the implementation of local structural measures.

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